

CSE 29 Week 2 Discussion

Bitwise Operators, Emojis, Project 1, and
more





Announcements

- Project 1 due Monday 10/12





Bitwise Operators

- << (left shift)
- >> (right shift)

Input 1	Input 2	& (“getter”)	(“setter”)	^
0	0	0	0	0
0	1	0	1	1
1	0	0	1	1
1	1	1	1	0



Unicode “Variations”

Demo

- `utf8-len()`
- ASCII vs 🦀 vs ☢



UTF-8 Error Cases

Example: 11111111

Code point ↔ UTF-8 conversion

First code point	Last code point	Byte 1	Byte 2	Byte 3	Byte 4
U+0000	U+007F	0yyyyzzzz			
U+0080	U+07FF	110xxxxyy			
U+0800	U+FFFF	1110wwwww	10xxxxxyy	10yyzzzz	
U+010000	U+10FFFF	11110uvvv	10vvvwwwww	10xxxxxyy	



Questions

- **Project 1**
- Lecture content: UTF-8, bitwise operators, hexadecimal, ...
- Labs: unix commands, git, vim, ...
- etc